

ST PETER-ORDING, Germany

WMO: 100280

Lat: 54.3278N

Lon: 8.6031E

Elev: 37

StdP: 14.68

Time Zone: 1.00 (EUC)

Period: 01-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	18.9	22.6	14.4	11.4	21.3	17.4	13.3	24.0	43.8	41.0	40.1	42.7	9.2	100	0.742

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.1	80.0	66.9	75.8	65.2	72.4	63.9	68.3	76.2	66.7	73.3	65.1	70.8	11.1	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.5	94.3	71.6	63.9	89.1	69.7	62.7	85.3	68.5	32.5	76.6	31.3	73.7	30.1	71.0	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 36.2	32.0	28.5	DB	15.1	86.5	5.8	2.6	11.0	88.4	7.6	90.0	4.3	91.4	0.1	93.4	(4)
(5)			WB	14.6	71.9	5.9	2.2	10.3	73.5	6.9	74.8	3.6	76.0	-0.7	77.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	49.1	35.2	34.7	39.2	47.4	53.9	59.4	64.1	64.2	59.2	50.6	43.9	36.8	(6)
(7)		DBStd	11.95	6.76	6.05	5.87	5.56	4.77	4.69	4.31	3.97	4.14	5.71	5.56	6.66	(7)
(8)		HDD50	2018	460	428	335	116	16	0	0	0	1	64	190	408	(8)
(9)		HDD65	5891	925	848	799	529	345	179	71	63	179	447	633	873	(9)
(10)		CDD50	1699	0	0	1	37	137	282	439	439	276	81	7	0	(10)
(11)		CDD65	98	0	0	0	0	1	12	44	37	4	0	0	0	(11)
(12)		CDH74	534	0	0	0	2	10	87	248	175	12	0	0	0	(12)
(13)		CDH80	97	0	0	0	0	0	8	54	35	0	0	0	0	(13)
(14)	Wind	WSAvg	14.3	16.0	14.6	14.5	13.1	13.8	14.5	13.6	13.4	14.3	14.2	14.8	14.5	(14)
(15)	Precipitation	PrecAvg	31.2	2.6	1.9	1.7	1.4	1.8	2.5	3.2	3.7	3.3	3.4	2.9	2.9	(15)
(16)		PrecMax	40.3	5.2	4.3	3.3	4.1	3.6	4.6	5.8	6.9	6.9	7.2	6.9	6.3	(16)
(17)		PrecMin	19.4	0.1	0.6	0.3	0.0	0.2	0.0	1.0	1.1	0.8	1.0	0.2	0.6	(17)
(18)		PrecStd	4.6	1.4	0.8	0.9	0.8	0.8	1.2	1.4	1.7	1.7	1.5	1.5	1.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.2	48.4	58.4	71.2	75.0	81.6	85.6	84.3	75.5	64.8	56.6	50.4	(19)
(20)			MCWB	47.2	45.7	48.9	55.9	61.4	65.8	68.6	67.7	64.5	60.2	54.2	48.3	(20)
(21)		2%	DB	46.9	46.0	52.6	65.6	70.5	76.4	80.6	79.1	71.7	61.9	54.0	48.0	(21)
(22)			MCWB	44.9	44.4	46.6	53.6	59.3	64.5	67.6	66.9	63.1	58.0	51.7	46.2	(22)
(23)		5%	DB	45.3	44.2	48.2	61.4	66.0	70.9	76.0	74.2	68.0	60.0	52.6	46.3	(23)
(24)			MCWB	43.8	42.7	44.7	52.5	56.5	62.4	66.0	65.4	61.7	56.7	50.4	44.6	(24)
(25)		10%	DB	43.9	42.6	45.9	56.6	62.4	67.0	72.3	70.8	65.4	58.3	51.0	44.9	(25)
(26)			MCWB	42.7	41.3	43.7	50.6	54.8	60.5	64.9	64.4	60.5	55.5	49.0	43.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.8	46.7	50.3	58.4	64.4	68.8	70.8	70.4	66.5	61.0	54.6	48.5	(27)
(28)			MCDB	48.9	47.5	54.9	68.6	73.7	78.0	80.8	78.8	72.6	64.6	56.1	49.9	(28)
(29)		2%	WB	45.5	44.6	47.4	55.1	60.1	65.8	68.8	67.7	64.0	58.9	52.2	46.5	(29)
(30)			MCDB	46.5	45.9	51.3	62.3	66.7	73.0	77.0	74.7	69.3	61.0	53.5	47.6	(30)
(31)		5%	WB	44.0	43.0	45.5	53.0	58.0	63.2	67.1	66.3	62.6	57.3	50.7	44.9	(31)
(32)			MCDB	45.0	43.9	48.0	59.7	63.8	69.1	74.4	72.3	66.8	59.4	52.2	45.9	(32)
(33)		10%	WB	42.7	41.4	44.0	50.9	56.1	61.0	65.3	64.9	61.3	55.8	49.3	43.7	(33)
(34)			MCDB	43.8	42.3	45.6	56.4	61.2	66.3	71.2	70.0	65.0	57.8	50.7	44.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.4	6.1	8.6	12.5	11.6	10.5	10.4	10.1	9.5	8.5	6.0	6.2	(35)
(36)			MCDBR	4.6	6.3	13.6	22.0	20.2	19.3	20.4	18.5	15.0	9.6	5.4	4.8	(36)
(37)			MCWBR	4.7	5.3	9.2	12.3	11.4	10.4	9.7	8.9	8.6	6.9	5.1	4.8	(37)
(38)		5% WB	MCDBR	4.6	5.2	11.7	18.5	16.5	16.6	17.1	14.4	12.0	8.4	5.2	4.9	(38)
(39)			MCWBR	4.8	4.9	8.5	11.1	10.4	9.9	9.2	8.3	7.8	6.9	5.1	5.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.328	0.346	0.372	0.405	0.397	0.395	0.411	0.408	0.387	0.364	0.341	0.319	(40)	
(41)		taud	2.291	2.320	2.284	2.248	2.334	2.362	2.332	2.348	2.392	2.397	2.336	2.288	(41)	
(42)		Ebn at Noon	193	233	252	260	270	271	263	257	246	224	187	169	(42)	
(43)		Edh at Noon	19	25	32	38	37	36	37	34	29	23	18	16	(43)	
(44)	All-Sky Solar Radiation	RadAvg	166	371	774	1291	1633	1695	1598	1303	931	503	207	121	(44)	
(45)		RadStd	20	53	70	145	164	140	193	100	104	57	24	9	(45)	

Historical Trends

		DBAvg		Heating		Cooling			Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (46 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-236	N/A	N/A	(47)

Nomenclature: See separate page